

## Research article

## Shaping the future of forestry in Germany: Stakeholder perspectives on optimizing forest management through Augmented Reality

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## ABSTRACT

Augmented Reality (AR) is revolutionizing various industries by enabling the real-time, context-driven integration of data into the physical environment. In forestry, AR offers opportunities to enhance operational efficiency, improve occupational safety, and support sustainable management practices. This study explored AR's potential in forestry through qualitative interviews with a diverse range of stakeholders, including forestry professionals and AR experts from multiple countries. The interviews, analyzed using qualitative content analysis, revealed a wide range of AR applications—spanning forest inventory, silvicultural operations, disaster management, logistics, and stakeholder communication. Key findings highlighted AR's ability to optimize processes such as real-time hazard detection, automated forest inventory, and interactive data visualization. The study also underscored AR's role in improving occupational safety, reducing errors, and supporting decision-making. Technologies such as Mixed Reality smart glasses integrated with GPS and Real-Time Kinematic systems were identified as essential for achieving the precision required. However, the study acknowledged barriers to implementation, including high acquisition costs, limited device durability, and user resistance due to discomfort and device complexity. Recommendations include the development of user-friendly applications featuring voice-controlled interfaces and modular visualization modes to improve usability and prevent information overload. Despite current limitations, AR holds potential to transform forestry by streamlining workflows, reducing environmental impact, and enhancing occupational safety. This research offers insights for AR software developers, hardware manufacturers, and forestry stakeholders, providing a roadmap for innovation and implementation. By overcoming identified barriers and leveraging AR's capabilities, forestry could achieve substantial improvements in efficiency and sustainability.

## 1. Introduction

Augmented Reality (AR) has emerged as a transformative technology across multiple industries, enabling real-time overlay of digital information onto the physical world. Unlike Virtual Reality (VR), which immerses users in entirely digital environments, AR enhances real-world settings with interactive, context-driven data (Carmigniani and Furht, 2011; Granshaw, 2020; Arena et al., 2022; Dargan et al., 2023). Over the last decade, AR has gained widespread adoption, particularly in fields such as manufacturing, healthcare, retail, and education (Azuma, 1997; Dangelmaier et al., 2005; Dirin and Laine, 2018; Kumar et al., 2020; Arena et al., 2022; Vigkos et al., 2022). Its ability to provide real-time assistance, streamline complex tasks, and offer hands-free solutions makes it invaluable in professional settings. Technological advancements in hardware, such as head-mounted displays and AR-enabled

smartphones and tablets (e.g., Microsoft HoloLens 2, iPhone, and iPad), have made AR more accessible and user-friendly. Despite these advances, AR faces challenges such as the high cost of devices and the need for highly skilled professionals to use them (Dangelmaier et al., 2005). Nonetheless, technology is expected to advance significantly, driven by increasing demand for remote collaboration and digital interactivity. The VR/AR industry in Europe has rapidly matured in recent years, with projections indicating it could reach a market value of EUR 766 billion by 2025 and contribute up to EUR 1.3 trillion to the global economy by 2030 (Vigkos et al., 2022). This growth is propelled by advances in both commercial and consumer VR/AR applications, underscoring the increasing relevance of these technologies in shaping future industries and economies (Vigkos et al., 2022).

In the environmental sciences, AR has begun to carve out a space, offering new avenues for visualizing data, supporting the management

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of natural disasters, and enhancing fieldwork (Veas et al., 2013; Haynes and Lange, 2016; Bravo Chávez, 2018; Khanal et al., 2022; Lamrabet et al., 2024). Within the forestry sector, in particular, AR is becoming an innovative tool to address some of the sector's most pressing challenges, such as labor shortages and climate-related forest risks (FAO, 2022; FAO, ILO & United Nations, 2023). The integration of this technology into the forest sector is highly relevant due to its economic, environmental, and societal significance in Europe. Forests cover 41 % of Europe's land area, contribute 0.7 % to GDP, and generate approximately 2.6 million jobs, while playing a vital role in biodiversity conservation, climate regulation, and hydrological cycles (Eurostat, 2021; European Environment Agency, 2024). Since most forests are actively managed, vast amounts of digital data are collected on infrastructure, ownership, inventory, and environmental conditions. Given their role in climate change mitigation, incorporating advanced technologies such as AR into forest management might become essential for enhancing sustainability and efficiency.

Traditionally, forest management has relied on manual data collection methods (e.g., field inventories) and spatial planning tools such as Geographic Information Systems (GIS). Recently, however, AR technologies have begun to complement these methods. While their implementation remains at an early stage, existing applications already illustrate the potential of AR to improve in situ spatial awareness, facilitate interactive visualization of geospatial data, and support real-time decision-making processes. The integration of AR into forestry initially centered around straightforward applications, primarily focused on educational purposes (Mund and Müller, 2019) or supporting various forest stakeholders (e.g., foresters, firefighters, harvester drivers) in conducting forest management-related activities that improve data visualization: logging measures (Nordlie and Till, 2015; Palonen et al., 2017; Sitompul and Wallmyr, 2019; Andersson, 2020), forest inventory processes (Ekenstedt, 2020; Tatsumi et al., 2023; Borz et al., 2024) and wildfire management (Vicens et al., 2006; Alférez et al., 2013; Bravo Chávez, 2018; Azpiroz et al., 2024). However, as the technology has matured, its uses have expanded to further areas, such as roundwood measurement (Borz et al., 2022) and natural disaster management (Vicens et al., 2006; Haynes and Lange, 2016; Haynes et al., 2018; Tomkins and Lange, 2019a, 2019b).

Building on these developments, the potential of AR in forestry is vast, offering numerous opportunities to support stakeholders in optimizing processes—understood here as field-based operational workflows such as forest harvesting or forest inventories—and enhancing occupational safety during forest operations. In this context, occupational safety refers to the protection of workers' health and safety throughout forest operations, including tasks such as tree felling, log extraction, and field-based inventory tasks. It is essential to focus development efforts on tailoring AR technologies to meet the specific needs of these stakeholders. In this regard, gathering insights from forest professionals and experts in AR software and hardware to better understand their requirements and evaluate how AR can address them is of paramount importance. However, the integration of AR in forestry still faces several challenges and limitations, some of which are common to the use of this technology in other application areas, while others are specific to the outdoor conditions such as those in the forestry sector (Avery et al., 2008). The former includes high acquisition costs for some AR devices and the need to hire highly skilled personnel, as previously mentioned. The latter involves challenges such as registration issues—specifically, the accuracy of AR projections under the forest canopy—as well as the requirement to ensure a stable network connection (Azuma, 2016; Fol et al., 2023).

Therefore, this social research study aimed to fulfill that task by analyzing the potential capabilities of AR in forestry and envisioning future scenarios for its implementation, based on the needs and insights of a wide spectrum of forest stakeholders and experts in AR. The findings provide crucial guidance for AR software developers and hardware manufacturers, highlighting pathways for further development, as well

as for researchers and forest stakeholders by showcasing how this technology can support them in conducting their activities. Moreover, this study identified a framework of conditions and barriers that must be considered during the development and deployment of AR in forestry. It is important to highlight that, while AR can be applied to various forestry domains—including urban forestry—this study specifically focused on its use in industrial forestry, forest management, and disaster response and infrastructure development.

To achieve this objective, this study was guided by the following key research questions:

1. What are the specific needs and expectations of forest stakeholders regarding AR technology?
2. How can AR contribute to optimizing forestry operations, improving efficiency, and enhancing occupational safety, as well as supporting other key areas such as disaster response and management, workforce training, forest infrastructure development, and machinery maintenance?
3. What are the key challenges, barriers, and framework conditions that need to be considered for the successful implementation of AR in forestry?
4. What future scenarios and development pathways can be envisioned for AR applications in forestry based on stakeholder insights?

By addressing these questions, this study aimed to provide a comprehensive understanding of AR's role in forestry and to establish a foundation for its effective development and adoption in the sector.

Finally, it is worth noting that this study was conducted within the framework of the *ForestAR* project, funded by the German Federal Ministry of Food and Agriculture (in German: *Bundesministerium für Ernährung und Landwirtschaft*—BMEL) through the Agency for Renewable Resources (in German: *Fachagentur Nachhaltende Rohstoffe*—FNR). The project focuses on a feasibility and potential study for the operational use of AR in forestry. It explores the application of AR technology—particularly the Microsoft HoloLens 2—to visualize digital forestry data directly in the field, enabling hands-free operation and enhancing both efficiency and safety in forestry tasks. The project aims to adapt and develop AR applications tailored to forestry conditions, thereby facilitating the digital transformation of the sector.

## 2. Materials and methods

### 2.1. Data acquisition through qualitative interviews

Among qualitative methods in social research, semi-structured interviews were selected for this study. This approach proved effective for exploring the subjective perceptions of identified stakeholders regarding relevant issues (Lamnek, 1995; Mayring, 2002). As indicated in the consulted bibliography (Witzel, 1985; Lamnek, 1995; Mayring, 2002), the interviews were conducted using a guideline that included four distinct sections:

1. Brief Introduction: the participants were asked to introduce themselves briefly and provide a concise overview of their current role, including details about their property or their company's products and services, as well as their experience with AR technology.
2. General probing concerning the interviewee's knowledge about digitalization and AR-technology, especially regarding the types of data used, hardware and software.
3. Specific probing, depending on the interviewee, concerning:
  - a. processes that could be optimized by using AR (e.g., tree felling, inventory, or planting)
  - b. framework conditions to consider when implementing and using AR in internal processes (i.e., internal organizational or operational workflows of forest-related stakeholders such as decision-making routines, or equipment planning)

- c. possible barriers, constraints and shortcomings that should be considered and addressed when implementing and using AR in forestry
  - d. further forest stakeholders which could benefit from using AR technology
4. Ad-hoc questions and interview conclusion, giving interviewees the opportunity to share their thoughts on the current study and suggest potential areas for future research.

The interviews were conducted in the period from August to November 2024 and recorded using Microsoft Teams and Zoom video conferencing software. They lasted between 12 and 57 min. A total of 47 participants were interviewed across 44 sessions. Interviewees were asked for consent to record the interviews and were informed about the data protection regulations as well as the guidelines on Safeguarding Academic Integrity of the University of Freiburg (Albert-Ludwigs-Universität Freiburg, 2024). In addition, they were informed about the storage duration of the interview recordings, which were scheduled for deletion once the analysis was completed. The content of the interviews was transcribed verbatim, without any additional commentary. The spoken language was captured exactly as expressed, including any deviations from standard written forms of German, Spanish, and English.

## 2.2. Selection of the interviewees

To gather the most relevant insights and expertise, a wide spectrum of German forest stakeholders was selected. Additionally, representatives from academic and research institutions, as well as companies specializing in AR software and hardware, were included. This latter group of interviewees comprised not only experts from Germany but also professionals from other countries (including selected forest stakeholders), chosen for their extensive experience and the valuable insights. The international participants were located in seven countries: Austria, Finland, Indonesia, Norway, Spain, Sweden, and Switzerland. The interviewee groups and their corresponding abbreviations are listed in Table 1 to assist readers in understanding the results and accurately attributing statements to the corresponding interviewees. In Table A1 in Appendix A, each group of interviewees is defined, and the number of interviewees in each group is indicated.

The identification of the interviewees was carried out through personal networks, the COST Action CA23135 - *Bringing Digital Data and Reality Together - Augmented Reality in Forestry (ARiF)*, and comprehensive internet research. Potential participants were contacted either by phone or by sending an interview request letter (via e-mail). During this initial contact, the *ForestAR* project was introduced, outlining its background and the study framework under which the interviews would take

place. Additionally, participants were asked to confirm their willingness to contribute to the study.

## 2.3. Evaluation of the interviews

The interviews were analyzed using qualitative content analysis based on the approach developed by Mayring (2022), specifically applying the technique of content structuring. This method enables a systematic evaluation of interview data according to defined criteria, thereby enhancing the interpretation of participants' responses. The analysis was conducted using the software MAXQDA 2022 (VERBI Software, 2022). The specific procedures employed throughout each phase of the analysis, and their application within the context of this study, are illustrated in Fig. 1, and the key content categories along with brief definitions are included in Table 2".

## 3. Results

This section is organized into four subsections that delve into the insights provided regarding the optimization of forest processes through AR integration—including aspects related to data, software and hardware; general conditions and key requirements; and the challenges and barriers to consider. It must be noted that the interviewees emphasized that the groups they represented would benefit from the use of AR technology. In addition, they identified further stakeholders that could also benefit, including energy providers (e.g., for maintenance work or for planning and constructing wind farms on forest land), farmers, local nature conservation authorities, landscape planners, and researchers (e.g., for identifying trial sites and collecting data at regular intervals, such as in the German National Forest Inventory).

### 3.1. AR integration in forestry: use cases and safety enhancements

The deployment of AR in forestry reveals significant benefits across various applications, which can enhance operational efficiency, safety, and environmental sustainability. A total of eight key application areas were identified including a total of 52 use cases in which AR can be integrated. The following lines outline use cases in the various identified areas of application. A selection of the most relevant use cases is illustrated in Fig. 2. The detailed potential applications identified by interviewees for these AR implementations can be found in Table B1 in Appendix B.

#### 1. Planning and inventory management

AR projection technology could offer significant support in forestry management and planning through a variety of applications. By reducing reliance on analog materials and enabling the simultaneous projection of all necessary information at the user's discretion, AR could improve the efficiency of forest management planning. This would be particularly important when landowners have small and widely scattered forest areas, or in agroforestry settings with many small plots and a lack of visible boundary markers [PrivFO, ResearchCom]. AR could display parcel boundaries or markers [ConsultCo], and could also delineate forest areas based on specific attributes such as ownership type, land use, contamination, pest or forest disease presence, archaeological sites, and protected habitats or tree groups [HeritageOff, StateFO, MuniFO]. Additionally, AR could project underground infrastructure—including abandoned mines, unexploded subsurface ordnance, and fiber optic cables—thereby enhancing both safety and planning accuracy [StateFO, PrivFO]. It could visualize LiDAR-derived point clouds within forest stands to support forest planning, for instance, by showing existing wood volumes and the volumes already extracted during previous harvesting operations [ForComp, TrainCtr].

By allowing stakeholders to visualize in advance how the landscape would appear following the implementation of specific measures, AR

**Table 1**  
Interviewee groups and their corresponding abbreviations

| Interviewee group                                                 | Abbr.       |
|-------------------------------------------------------------------|-------------|
| State forest ownership                                            | StateFO     |
| Municipal forest ownership                                        | MuniFO      |
| Private forest ownership                                          | PrivFO      |
| Forest management association (forest owners' cooperative)        | ForCoop     |
| Forest management company                                         | ForComp     |
| Forestry consulting company                                       | ConsultCo   |
| Timber transport company                                          | TimTransp   |
| Wood industry company                                             | WoodInd     |
| Law enforcement and public safety authorities                     | SafetyAuth  |
| Nature conservation authorities and organizations                 | NatConOrg   |
| Federal/regional offices for archaeological heritage conservation | HeritageOff |
| Social insurance and occupational safety organization             | SocIns      |
| Academic and research community                                   | ResearchCom |
| Forestry training center                                          | TrainCtr    |
| AR software companies                                             | ARSoft      |
| AR hardware companies                                             | ARHard      |

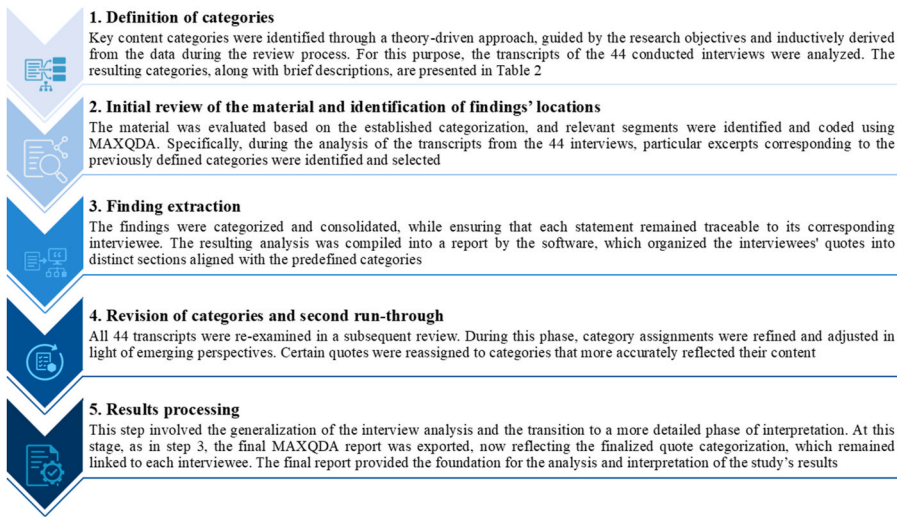


Fig. 1. Phases and procedures of the analytical process employed in the study

Table 2  
Definitions of the key content categories

| Category                                 | Definition                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Potential users/forest stakeholders      | Identification of individuals or groups who are directly or indirectly involved in forest management and who may benefit from or interact with AR technologies in forestry contexts beyond the interviewees groups considered in this study                                                                                     |
| Potential applications of AR in forestry | Identification and description of the potential use cases, scenarios, processes, and practical implementations of AR within the forestry sector, including operations or tasks such as planning, monitoring, training, and education, which could be optimized or contribute to improved occupational safety for forest workers |
| General conditions                       | Identification and description of the overarching environmental, institutional, or contextual factors that influence the implementation or feasibility of AR solutions in forestry settings, as well as the conditions under which forest stakeholders would adopt AR in their activities and internal processes                |
| Requirements                             | Identification and description of the necessary preconditions, resources, or criteria—technical, operational, or organizational—that must be met for the successful deployment and utilization of AR technologies in forestry, as perceived by the interviewees                                                                 |
| Barriers and limitations                 | Identification and description of the challenges, constraints, or obstacles that may hinder or limit the adoption, effectiveness, or scalability of AR applications in forestry                                                                                                                                                 |
| Previous knowledge or experience with AR | Description of the extent to which participants or stakeholders were familiar with or had prior experience with AR before the study, including how this background may have influenced their perceptions and expectations, as well as a description of their previous experience and knowledge                                  |
| Data to be displayed                     | Identification and description of the specific types of spatial, visual, or textual information intended to be projected through AR to support forestry-related activities                                                                                                                                                      |
| Hardware to be used                      | Identification and description of the physical devices and equipment required to operate AR systems in forestry applications                                                                                                                                                                                                    |
| Software to be applied                   | Identification and description of the applications, platforms, or programming environments that may be used—or have already been used—to develop, manage, or execute AR functionalities in forestry-related use cases                                                                                                           |

could enable them to determine whether a measure should proceed or if an alternative course of action would be preferable [ResearchCom, PrivFO]. This capability could help anticipate the outcome of work and prevent errors, thereby improving overall efficiency [ResearchCom, MuniFO]. For navigation purposes, forest stakeholders could use AR to visualize which forest roads to use, as well as assess their condition—optimizing logistics and enhancing safety [StateFO, MuniFO, PrivFO, ConsultCo, ForComp, TimTransp]. This use case has already been developed and evaluated using the hardware and software provided by the company V-Labs (V-Labs SA, Plan-les-Ouates, Switzerland), as part of the research study conducted by Purfürst and De Miguel-Díez (2024) (see Fig. 3).

It was found that AR could be used to project the development and future appearance of forest stands based on selected species and management practices, thereby supporting consensus among forest owners, planners, and forestry companies regarding silvicultural measures [TrainCtr]. Interactive AR projections might also display relevant guidelines, regulations, laws, or funding program details, offering benefit estimations for proposed decisions—particularly for forest owners in regions of Germany heavily affected by forest pests such as bark beetles, where reforestation initiatives are currently being planned [PrivFo]. In addition, AR might facilitate civic engagement and participation—especially among younger audiences—to raise awareness of forestry operations conducted for the purposes of forest health and sustainability [ResearchCom, ConsultCo].

AR projections might provide key stand information such as wood volume and expected benefits [PrivFO] and could document completed and planned measures for record-keeping purposes [ForComp, PrivFO]. Production and efficiency could be increased, as detailed information regarding forest stands might be collected digitally in the field—replacing traditional practices such as handwritten notes, saving georeferenced points, or taking photographs [ConsultCo, PrivFO]. Furthermore, AR could enhance data collection by projecting sample plot centers, boundaries, and surveyed sample areas, thereby optimizing forest inventory tasks. For instance, forest workers would no longer need to measure the radius of a circular plot to determine whether a specific tree should be included and measured, thus increasing efficiency [ARSoft, ConsultCo]. This optimization might offer a more efficient alternative to commonly used tools [StateFO, ARSoft, ConsultCo, ForComp]. For data validation, AR might bring satellite or airborne laser scanning (ALS) LiDAR data into the field for real-time comparison [ForComp, ResearchCom], and could digitally mark trees that have been processed—such as delimbed trees intended for high-quality wood production—enhancing both tracking and planning [ForComp,



Fig. 2. Overview of potential AR applications in forestry



Fig. 3. Example of AR-supported forest road navigation for improved logistical planning and safety

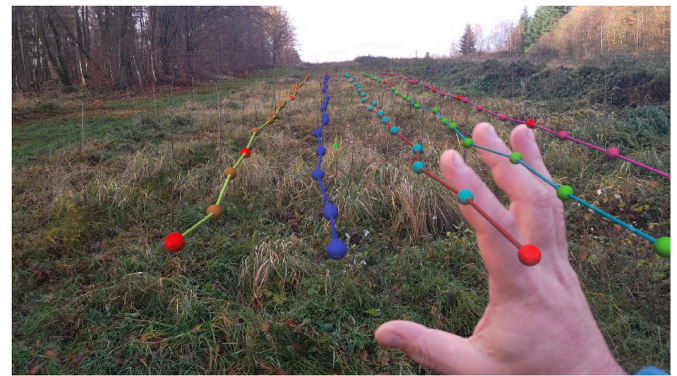


Fig. 4. AR projection of planting lines and seedling positions

StateFO]. Lastly, in agroforestry, landowners or planners might find it easier to locate plots and visualize production using AR, thereby optimizing management and planning [ResearchCom].

## 2. Silvicultural, harvesting and reforestation operations

AR projection technology could provide crucial support for forestry operations by enhancing safety, efficiency, and precision. For instance, AR might project guidelines for silvicultural treatments to demonstrate procedures such as proper tree planting or felling techniques for untrained personnel [PrivFO]. In addition, AR could assist in visualizing special treatments required for certain tree species—for example, eucalyptus regrowth under optimal conditions—or for improving productivity in agroforestry systems [WoodInd, ResearchCom]. AR might also be used to project specific planting locations or planting lines for seedlings, using different symbolologies for each tree species. This would enable precise and organized planting, not only in forested areas but also in agroforestry plots [StateFO, ForComp, ResearchCom]. This use case was developed and evaluated in the context of the *ForestAR* project, utilizing hardware and software provided by V-Labs (V-Labs SA, Planles-Ouates, Switzerland) (see Fig. 4).

During thinning operations, AR might display work instructions by projecting skid trails, indicating their locations and distances to the next trail, along with marking trees designated for felling. It could also categorize different parts of the tree by quality, assisting forest workers in identifying which trees should be felled—even when visibility is

reduced due to tall, dense under-canopy vegetation or regeneration [StateFO, ForComp, MuniFO, PrivFO, ConsultCo].

In harvesting operations involving chainsaws and machinery, AR could be used to display in the forest decision-support tables that assist forest workers in selecting appropriate tools or equipment based on the weight and structural characteristics of the tree. For example, in Germany, reference tools such as the “Calmbacher Table” (in German: “*Calmbacher Tabelle*”) provide estimates of the pulling force required to safely fell a tree, taking into account variables such as diameter at breast height (DBH), species group (hardwood or softwood), and anchor height. Projecting such information in the field via AR could support safer and more efficient work practices [TrainCtr]. For additional support, vertical lines or 3D geometric figures could be displayed in AR to assess tree conditions and inclination, identify appropriate cutting techniques, and suggest the optimal felling direction [ForComp, TrainCtr]. Projections of planned operations across designated areas could enable personnel to work concurrently in multiple locations by allowing them to visualize which areas had already been completed by other teams. This would help avoid redundant work and save time, as teams would no longer need to travel to or repeat tasks in locations where operations had already been carried out. This visualization could also assist foremen in monitoring progress and help replacement personnel understand which tasks had been completed and which still required attention. [ConsultCo, StateFO].

AR projections might identify trees with diameters exceeding 75 cm, which certain wood industry companies are unable to process due to technical limitations [WoodInd]. They could also display the specific

tree sections to be cut, organized by assortment type [WoodInd].

Key expected benefits of AR implementation could include both quantitative and qualitative improvements in timber production [WoodInd], as well as the optimization of forestry processes by eliminating the need for ranging rods and costly spray paint for tree marking. Instead, AR would enable digital tree marking, allowing for easier identification by harvester operators. This transition away from spray paint might not only reduce exposure to toxic components but also help avoid issues with fading markings, which typically deteriorate within two to three years [TrainCtr]. Additionally, AR projections might enhance productivity by enabling the simultaneous treatment of multiple forest areas, reducing occupational accident risks, and promoting a safer overall work environment [ConsultCo, TrainCtr].

### 3. Logistic operations

AR projection technology might offer substantial support in forestry logistics—referring here to the set of operational activities involved in the post-harvest handling of timber, including the extraction and roadside stacking of logs in forest areas, as well as their subsequent transportation by truck to the production facilities. This could enhance location tracking, route navigation, and operational management across the forestry logistics chain. It could assist forwarder operators by projecting the exact location of logs after processing, enabling faster and more efficient retrieval [WoodInd]. Additionally, AR could project essential documents commonly needed in the field—such as wood lists, contract documents, work orders, forest management plans, and nature conservation documents—thereby reducing reliance on physical paperwork [ForComp]. To streamline forestry logistics—particularly the transport and handling of roundwood—AR projections might display numbered log stacks along with their designated destinations or purchasers. Additionally, scanned wood stacks could be monitored through point cloud data to detect any potential roundwood theft [StateFO, MuniFO, PrivFO, ForComp, TrainCtr, WoodInd]. Moreover, AR projections could provide drivers with detailed load-related information, including weight, stacking methods, and loading recommendations, thereby supporting optimal loading practices [ResearchCom]. The expected benefits would include improved logistical efficiency through enhanced, real-time information for drivers, which would also contribute to reduced accident risks. This approach would optimize situational awareness, enabling better decision-making and safer navigation. Furthermore, this technology would be expected to improve fuel efficiency and contribute to a reduction in CO<sub>2</sub> emissions, thereby supporting the environmental sustainability of forestry operations [ResearchCom].

### 4. Machinery maintenance and manufacturing

AR technology could provide valuable support for the maintenance and repair of forestry machinery by displaying detailed procedures for proper upkeep and necessary repairs [WoodInd, TrainCtr, ForComp, PrivFO]. Additionally, AR could assist in quality assurance during the manufacturing of forestry machinery, helping to ensure that high standards are consistently met throughout the production process [ResearchCom]. This technology could reduce the need for specialized staff training, thereby improving both economic and human resource efficiency within companies [WoodInd, ForComp, TimTransp].

### 5. Monitoring and conservation measures

It was suggested that AR could support forest conservation management by providing visual verification of planned activities, ensuring that forest operations—such as forest plantations—are carried out as intended [ConsultCo, StateFO]. AR could also display the development stages of a forest stand, enabling managers to monitor growth and assess long-term outcomes [ConsultCo]. Additionally, AR projections might

mark planting locations to verify the success of reforestation efforts, ensuring that planting operations align with ecological and management objectives [ConsultCo]. Through AR, habitat tree groups could be tagged to support the protection of essential biodiversity [MuniFO]. Areas affected by pests or diseases could be identified and marked, aiding in the targeted management and ecological restoration of impacted regions [PrivFO, TrainCtr]. AR technology could enhance forest management by enabling real-time tracking of activities, improving oversight and quality control, and facilitating prompt corrective actions. It could support strategic decision-making through detailed visualization of forest growth and resource needs, thereby aiding sustainable management practices. AR could also promote operational efficiency by streamlining reporting processes and verifying planting success, which would reduce manual labor and highlight areas requiring follow-up actions. This, in turn, could promote higher survival rates for newly planted trees and optimize resource allocation. Additionally, the digital tagging of habitat tree groups could support biodiversity conservation by enabling easy on-site recognition of tagged trees that host wildlife—including protected species—that may not be felled. Unlike traditional spray markings, which fade within three to four years and may leave these habitats vulnerable during future harvesting operations, digital tags remain visible over time, thereby ensuring long-term protection of critical habitats [MuniFO].

### 6. Rural and forest infrastructure development

The AR projection of essential information could optimize forest road repair processes by accurately highlighting the required construction materials at specific locations and identifying the necessary machinery [ARSoft, ResearchCom, ForComp]. Additionally, AR could support the projection of dam, check dam, and fish ladder construction projects in forested areas using Building Information Modeling (BIM) format, enabling on-site visualization and facilitating any necessary adjustments [ResearchCom]. This capability could enhance project planning, streamline decision-making, and optimize the execution of infrastructure development in forest areas [StateFO, ForComp, ResearchCom].

### 7. Natural disaster management and risk prevention in forest areas

AR projection technology could provide essential support in wildfire suppression and emergency response by continuously displaying the location of firefighting teams and escape routes, thereby enhancing orientation and situational awareness in the field [ConsultCo, ResearchCom, SafetyAuth]. It could enable real-time projection of hot-spot locations, aiding in efficient and targeted suppression efforts [ResearchCom]. Additionally, AR might simulate wildfire spread based on current weather conditions and vegetation characteristics, enabling firefighters to anticipate fire behavior and optimize their wildfire suppression strategies [ResearchCom, SafetyAuth]. Projections could also display various vegetation fuel models and forest types, providing firefighters with critical insights into fuel sources and the potential for fire spread [SafetyAuth]. Relevant climatic data—such as ambient temperature, relative humidity, wind speed, and wind direction—might be projected to inform tactical decisions during wildfire suppression operations [SafetyAuth]. Furthermore, AR could enhance logistical coordination by marking the condition of forest roads and identifying essential points of interest, including water intake and rescue locations [SafetyAuth]. The use of AR in these contexts could optimize wildfire suppression strategies, helping to prevent accidents, injuries, and even fatalities among firefighting teams [ConsultCo]. With real-time definition and adaptation of suppression tactics, AR could enhance operational efficiency while safeguarding personnel through dynamic, data-informed decision-making [ResearchCom]. Beyond wildfire response, AR might also mark flood-prone areas and active flooded zones, supporting broader disaster preparedness and environmental risk management efforts [ResearchCom].

## 8. Capacity building in forestry

In the framework of capacity-building in forestry, the application of AR could offer significant advantages by visualizing successive stages of forestry operations [ResearchCom], illustrating step-by-step procedures for activities such as silvicultural measures [WoodInd, ForComp], and simulating wildfire scenarios to effectively train firefighting units in both indoor and outdoor environments [SafetyAuth]. Expected benefits could include minimizing the learning curve for new forestry professionals [ResearchCom, WoodInd] and providing a valuable demonstration for students on the effects of implemented silvicultural measures, as well as on the evolution and growth of forest stands—for instance, in marteloscopes [ConsultCo]. Additionally, AR projections might model the progression of forest stands and monitor completed interventions, such as logging operations, while enabling the visualization of upcoming measures and the anticipated state of forest areas post-logging [ConsultCo].

In addition to the integration of AR in the aforementioned forestry areas, numerous interviewees identified additional ways in which AR could enhance occupational safety during forest operations. These suggestions are outlined in the following four subsections and illustrated in Fig. 5.

### 1. Real-Time monitoring and position tracking

AR projection technology could enable team leaders to monitor the real-time positions of both personnel and forestry machinery, enhancing the ability to track work progress in the field and respond quickly to accidents [TrainCtr, PrivFO]. AR could also facilitate the connection between forestry workers by displaying their respective locations within the forest and highlighting any occupational hazards or risks present [StateFO, SocIns]. Furthermore, AR projections might be used to display each worker's vital signs—such as hydration levels and heart rate—visible only to the individual to ensure data protection [SocIns]. Additionally, AR could project designated rescue points within forest areas, aiding in emergency response planning [ForComp, MuniFO].

### 2. Identification of major hazards

AR projection technology could provide forestry workers with essential safety insights by visualizing major hazards and deadwood within a forest stand before beginning their work [StateFO, TrainCtr, ForComp, MuniFO]. It could also highlight hazardous areas resulting from terrain particularities, such as small ravines and steep slopes,

allowing workers to navigate with enhanced awareness [TrainCtr]. Additionally, AR projections might identify forest areas impacted by pests or diseases that pose risks to workers' health [TrainCtr].

### 3. Safety in harvesting and logistics operations

AR projection technology could greatly enhance safety and precision in forest operations. For example, AR could project the felling direction and designated felling zone for a tree—taking into account different tree species—along with a warning function to alert workers if someone enters this area during motorized saw operations [TrainCtr, ForComp, SocIns]. Additionally, it might display the mandatory 20-m safety distance, even when using remote-controlled felling wedges [SocIns].

The projection of specific cutting techniques for motor saws, including the handling of storm-damaged timber, might help visualize tension zones (compression and tension) and focal points within a tree, along with guidance for holding the chainsaw perpendicular to the tree during felling [TrainCtr, SocIns]. AR could also assist by displaying stump diameter, undercut, and hinge dimensions, supporting safe cutting practices.

Furthermore, AR could also enhance safety and situational awareness for drivers of trucks or heavy machinery by projecting points of interest, infrastructure details (such as road curves), recommended speeds, forest road conditions, and alerts for nearby vehicles or obstacles—thereby facilitating safer driving and maneuvering within forested areas [ResearchCom].

For training purposes, AR guidance on optimal cutting depth might help avoid errors, such as cutting through the hinge, and could trigger warnings if the guide bar is inserted too deeply [TrainCtr]. Moreover, AR projections might display wind direction and speed, with an automatic halt notification when wind speeds exceed 49 km/h, ensuring safe felling conditions [SocIns].

Numbered skid roads could also be visualized, guiding forest workers around active machinery zones to prevent work accidents [TrainCtr]. For timber extraction using a winch, AR might display the winch line and necessary safety distance, issuing warnings if the line is improperly positioned. Real-time AR projections of the pulling force exerted by the forestry winch could help operators monitor cable load and related safety factors [SocIns].

Additionally, AR technology might alert forest personnel when heavy machinery is nearby, enhancing situational awareness and reducing the risk of accidents. This alert system could be enabled by equipping heavy machinery with GPS devices and geofencing capabilities [ARSoft, TrainCtr].

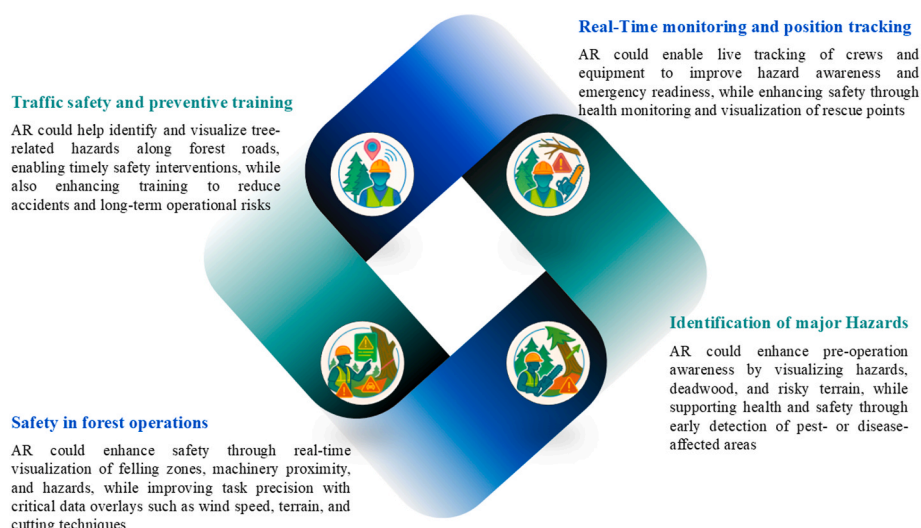


Fig. 5. Overview of AR-supported occupational safety applications in forestry

#### 4. Traffic safety and preventive training

In the context of traffic safety, AR projection technology could display, directly on affected trees, the identified risks—such as deadwood in the crown, structural instability, or leaning towards areas frequented by people or machinery (e.g., a forest road or a parking area)—along with the specific silvicultural or technical measures required to restore safety (e.g., removal of dead branches, crown securing, or felling). This would replace conventional spray markings, enhance efficiency during field operations, and improve occupational safety [TrainCtr, FMO, PrivFO]. Furthermore, AR-based training could be instrumental in accident prevention by equipping forestry workers with the knowledge and skills needed to enhance safety and reduce risks in future operations [ResearchCom].

#### 3.2. Data, hardware, and software used for AR integration in forestry

Interviewees, drawing on their experience and interest in leveraging potential AR applications in forestry, indicated several preferred data formats. These would include text, numerical data, LiDAR data, CAD data, and simulation data [ResearchCom, StateFO, PrivFO, ForComp, ConsultCo, TimTransp]. Additionally, they valued the use of widely applied reference tables, such as the aforementioned Calmbacher table [TrainCtr]. Other preferred data types would include geospatial data, such as georeferenced points, boundary markers, and geometric figures for visualizing skid trails or navigating through forest areas (as shown in Fig. 3). Interviewees also emphasized the importance of using AR to project relevant distances and measured parameters, such as the distance to the next skid trail, tree diameters at breast height (DBH), and tree heights [ForComp, PrivFO, WoodInd, TimTransp, ConsultCo, SafetyAuth].

Given the absence of a standardized framework for AR, ensuring compatibility across data, software, and hardware is crucial to enable developers and users to efficiently create and utilize AR [ResearchCom, ConsultCo]. Therefore, AR software developers would need to prioritize compatibility among various 3D data formats and associated tools. One widely adopted format is the GL Transmission Format (glTF), which is favored for its optimized performance in real-time applications. As a result, developers frequently convert other common 3D formats—such as the Wavefront OBJ (primarily used for storing geometric data) and Filmbox (FBX) (renowned for handling complex animation data)—into glTF to maintain a streamlined workflow. Another critical format in the AR ecosystem is Universal Scene Description (USD). USD provides a flexible framework for representing complex scenes and supports interoperability between diverse software platforms. In addition, when dealing with building and construction data, the Industry Foundation Classes (IFC) format remains the most widely adopted standard.

While the use of tablets and smartphones to visualize data in AR is widely spread, a broad spectrum of smart glasses or Mixed Reality (MR) glasses have been—and continue to be—considered for use in forestry. Several devices are currently being evaluated, including the Magic Leap 2, Microsoft HoloLens 2, Apple Vision Pro, Meta Quest 3, and Varjo XR-4 [ARSoft, ARHard, ResearchCom]. The results of these tests have revealed the limitations of several devices—some of which are insurmountable [ARSoft, StateFO]. Key limitations that reduce the spectrum of potential applications include, for example, the need to remain constantly connected to a laptop via a cable of fixed length. This requirement has prevented certain devices, such as the Varjo XR-4, from being used in several of the aforementioned forestry applications [ARHard]. Conversely, the Varjo XR-4 would be highly suitable in terms of data protection, as its operational design ensures strong data privacy safeguards [ARHard]. In instances where the use of smart glasses has not proven effective, these devices are replaced by smartphones or tablets, which provide a more affordable—and in some cases, even more efficient—means for users to project the necessary data in AR [ResearchCom, StateFO]. Additionally, cost-effective devices such as cameras (e.

g., GoPro, DJI) can be used to display the desired information in AR through video overlays [ConsultCo]. In contrast, the future potential of other devices—such as the Meta Quest 3—was highlighted in the interview conducted with an AR software developer. This device might be preferred due to its compatibility with WebXR and its relatively low cost [ARSoft]. However, the Meta Quest 3, classified as passthrough-type smart glasses, is not considered suitable for forestry applications by several stakeholders due to potential occupational safety risks [StateFO, ARHard]. These risks stem from the fact that users view their surroundings through a camera-supported display rather than directly, as they would with see-through devices like the Microsoft HoloLens 2 [StateFO]. Consequently, upcoming AR glasses—such as Orion, developed by Meta—are currently regarded as promising alternatives [ResearchCom].

The use of smart glasses is closely linked to the type of data to be projected and the software employed. For instance, some applications can be developed specifically for use with the Microsoft HoloLens 2, utilizing Microsoft's proprietary software. Others are designed to work with web-based platforms, alternative software, and specific smart glasses, such as the Magic Leap 2 or Varjo XR-4 [ARSoft, ARHard]. Some companies have already developed solutions that integrate MR smart glasses with protective helmets used in work areas and Real-Time Kinematics (RTK) technology. A GPS receiver with RTK capability can process standard signals from Global Navigation Satellite Systems (GNSS) in conjunction with a correction stream, enabling it to achieve positional accuracy and thereby improve the accuracy of the projected geodata [ARSoft]. The use of certain MR smart glasses allow users to capture data that can later be projected, such as photogrammetry-derived 3D models or LiDAR-based point clouds. For these purposes, it is essential to ensure the high quality of the embedded hardware and sensor technology [ARHard, ResearchCom].

In general, several interviewees identified two major game development platforms used in the creation of AR solutions: Unity and Unreal Engine [ResearchCom]. With regard to software development, developers strive to ensure that their solutions comply with the requirements of the respective platforms and proprietary standards, such as WebXR, ARKit and ARCore [ResearchCom, ARSoft]. WebXR is a web-based Application Programming Interface (API) that enables developers to create immersive AR and VR experiences directly within web browsers, thereby broadening access to these technologies without the need for native applications. Similarly, ARKit and ARCore—proprietary development kits offered by Apple and Google, respectively—are Software Development Kits (SDKs) that provide developers with the essential tools and libraries needed to create AR applications. ARKit provides a robust framework for building AR experiences on iOS devices, while ARCore offers comparable tools for Android devices. By adhering to these standards, developers can enhance performance, ensure cross-platform compatibility, and deliver a more consistent user experience across different hardware environments. The key difference between WebXR and the aforementioned SDKs is that WebXR is a browser-based API, whereas the SDKs represent more traditional tools for developing native applications, specifically for Android or iOS devices. Apart from the aforementioned ARKit and ARCore, the most common SDKs include Vuforia and Unity AR Foundation. An additional software tool used for creating 3D models is Speedtree [ResearchCom, ARSoft].

Key advantages of using WebXR include its direct access via web browsers, eliminating the need for app downloads, and cross-platform compatibility, allowing a single development to function across multiple devices, such as smartphones, desktops, and AR/VR headsets. WebXR also simplifies deployment, as updates can be pushed online without user intervention, thereby reducing both development costs and the need for multiple platform-specific versions. Additionally, its open, shareable nature via URLs enhances its accessibility, and as an open standard, it fosters interoperability and flexibility, avoiding dependence on proprietary platforms. Interviewees reported promising developments in AR applications using WebXR, including space tracking

and Gaussian splatting, which are currently underway. These advancements involve scanning environments and marking georeferenced points of interest. As a result, the scanned environments are transformed into “smart spaces” applicable both indoor and outdoor contexts [ARSoft, ResearchCom].

### 3.3. General conditions and key requirements to implement and use AR systems across distinct types of forest stakeholders

According to the interviewees, several factors are crucial in this decision-making process regarding the implementation and use of AR in forestry: 1) financial capacity, 2) affinity for technology, 3) level of professionalization of workers, 4) size of the property (in the case of forest owners), 5) age of the workforce, 6) forest management objectives or the functional roles of different forest types, and lastly, 7) the acceptability of this technology. These factors may vary depending on the type of stakeholder involved—particularly regarding forest owners. For larger or publicly owned properties (whether state or communal), personnel are more likely to have a higher degree of professionalization, and the owners themselves typically have greater financial resources to adopt AR technology. Affinity for digital tools often depends on the individual using them, with age being a significant factor; younger workers generally demonstrate greater adaptability to digital technologies and, consequently, to AR systems. In the case of smaller forest properties, the owner is more likely to be advised by public administration bodies or forestry associations (“forstwirtschaftliche Zusammenschlüsse”) [PrivFO, ConsultCo, ForComp]. These two stakeholder groups can either utilize AR technology to advise forest owners or directly manage the forest. In private entities—such as industry, forestry companies, or large private forest holdings—forestry management objectives may influence the adoption of this technology, sometimes outweighing financial considerations, as the optimization potential is widely recognized. In public forests, where nature protection and biodiversity conservation are key priorities, AR technology that supports these goals may significantly increase the likelihood of successful implementation. Acceptability is a critical factor in the adoption and effective use of AR. First, it must demonstrate reliability and sufficient accuracy. Second, it must provide stakeholders with clear added value and the specific, context-relevant information they need on-site in order to ensure successful implementation [StateFO, PrivFO].

The respondents emphasized that essential conditions for the adoption of this technology include its user-friendliness, small and lightweight design, and sufficient robustness to withstand the demands of forestry work and extreme weather conditions—such as freezing or very high temperatures, fog, humidity, or dust. Furthermore, the hardware must provide application-specific accuracy and incorporate a design that minimizes any risk of workplace accidents [StateFO, PrivFO, ForComp, ResearchCom]. Additionally, an offline version of the applications should also be made available to ensure functionality in forested areas where internet connection may be limited [SocIns]. Some interviewees specified varying accuracy requirements depending on the specific application. For example, an accuracy of 10 cm was considered necessary for displaying forest boundaries, 50 cm for showing the location of wood stacks—particularly when multiple stacks are placed close together or when labeling is unclear due to a use of dark colors or weathered spray paint—and a maximum of 1 m was deemed acceptable for displaying individual trees [PrivFO, ForComp]. Furthermore, device and hardware maintenance should not be particularly time-consuming. At present, smart glasses primarily require regular software updates—which can typically be managed automatically through the associated app store—and basic routine cleaning [ARSoft, PrivFO, ForComp]. Additionally, effective solutions must address the challenges and barriers outlined in the following section. AR smart glasses should be incorporated into the personal protective equipment (PPE) without requiring cable connection, instead utilizing wireless technologies such as Bluetooth. Ideally, the device should be integrated into the safety

helmet, along with hearing protection and either protective eyewear or a shield positioned in front of the smart glasses to guard against hazards such as stones and branches during forestry operations. To enhance projection accuracy and precision, additional components—such as RTK technology—should also be included [ARSoft, ForComp, PrivFO, StateFO, ARHard]. Other important factors to consider include data currency and the option to select data for display, ideally through voice selection [StateFO, MuniFO, PrivFO, SafetyAuth]. Data currency is particularly important for various stakeholders when planning forest inventories, identifying hazardous areas due to deadwood, implementing silvicultural measures, managing timber transactions, or coordinating temporary storage of wood in the forest. In certain situations, such as wildfire suppression, real-time information may be required [ForComp, ConsultCo, StateFO, PrivFO, SafetyAuth]. Moreover, enabling users to select which data are displayed is essential to simplify operation, reduce distractions, and minimize the risk of motion sickness caused by unnecessary or excessive information [StateFO, ForComp, SafetyAuth, ResearchCom].

### 3.4. Challenges and barriers to the implementation of AR in forestry

The interviewees reported several challenges that may arise when adopting AR in forestry, which can be categorized into technical (11), financial (2), organizational (3), legal (1) and human-related (4) barriers. Insights from interviews revealed that each category presents specific obstacles that must be addressed to ensure successful AR integration in forestry. The next lines outline the barriers and limitations identified by interviewees for implementing AR in forestry. A detailed description of those limitations can be read in [Table C1](#) in [Appendix C](#).

- Technical barriers involve device limitations, such as insufficient water and dust resistance, lack of durability, and issues related to weight, battery life, and sensitivity to lighting conditions. Smart glasses are often fragile and may be at risk of damage in rugged forest environments. Additionally, a reduced field of vision and latency problems can negatively affect the user experience and compromise the accuracy of georeferenced data projection—particularly under dense canopy cover. Furthermore, technical requirements like the need for stable internet connectivity in remote forested areas present additional challenges for effective AR deployment [ResearchCom, ARSoft, ForComp, SocIns, FMO, TimTransp, TrainCtr, PrivFO].
- Financial barriers are primarily related to the high costs associated with the acquisition and development of AR devices and applications, as well as the licensing fees. These challenges are particularly pronounced when the technology is available only for individual users, making large-scale or shared implementation less feasible [StateFO, ResearchCom, TrainCtr, ConsultCo].
- Organizational barriers include the limited level of digitalization within internal processes, which hinders the effective integration of AR. Additionally, accessibility is constrained by restrictions set by device manufacturers such as Apple and Microsoft, whose platforms may limit application development. However, new EU regulations—such as the Digital Markets Act (DMA)—are expected to address some of these limitations in Europe. This legislation, aimed at promoting competition and reducing monopolistic practices, will require companies like Apple and Microsoft to permit sideloading, allowing users to download applications from third-party sources rather than being restricted to official app stores [ResearchCom, ARSoft].
- Legal barriers involve compliance with data protection laws and national regulations that may restrict the use of AR originating from third countries. In addition, occupational safety regulations play a critical role in evaluating whether AR can be safely used in forestry operations, potentially limiting their adoption until proven compliant with national safety standards [ARSoft, ResearchCom].

- Human-related barriers include varying levels of staff affinity and comfort with digital tools. For AR to be successfully adopted, personnel must receive adequate training—especially for features such as speech recognition, which require familiarization and proper usage. Additionally, equipment such as smart glasses may cause physical discomfort due to their bulk or fit, and the over-projection of information can lead to distraction, increasing the risk of accidents in forest environments [ResearchCom, ARSoft, ForComp, TimTransp].

#### 4. Discussion

##### 4.1. Stakeholder-informed pathway for the implementation of AR in forestry

AR technology has already demonstrated promising potential in the forestry sector, with applications already developed and implemented for forest inventory and harvesting operations (Vicens et al., 2006; Alférez et al., 2013; Arboreal AB, 2018; Bravo Chávez, 2018; Balindt, 2020; Ekenstedt, 2020; Arboreal AB, 2022; Borz et al., 2022; Niță and Borz, 2023). However, these initial successes only scratch the surface of AR's full capabilities. The findings of this study revealed that forestry stakeholders have a wide variety of needs and use cases. Across the interviews, it became clear that AR holds immense potential for optimizing forestry processes, streamlining operations, and significantly enhancing occupational safety. Consequently, developments in this field could present a significant opportunity for software developers and device manufacturers to innovate and respond to sector-specific needs. The successful implementation of AR would depend not only on a clear demonstration of its benefits, but also on the stakeholders' willingness to adapt existing processes. This adaptation would involve advancing digitalization efforts to ensure that the necessary data are available and suitable for AR visualization [ResearchCom].

An implementation pathway for AR in forestry was deduced based on the statements of several interviewees [StateFO, PrivFO, ConsultCo, TrainCtr, ResearchCom] (see Fig. 6). It would encompass three phases: initial implementation, expansion and validation, and ultimately, long-term adoption. The first phase would begin with an assessment of the forestry sector to establish a baseline of technological readiness and contextual constraints. The availability of the data needed for projection using AR systems should also be carefully examined, since this is essential for implementing this technology. Equally important are the levels of digitalization and the technological affinity of the forest stakeholders. In addition, that evaluation should include the aspect regarding the treatment of sensitive company or staff information. Several

interviewees highlighted the importance of keeping internal data strictly within the company's domain [PrivFO, ARSoft]. A pilot project could then explore AR applications such as forest navigation or the visualization of forest boundaries within selected processes, e.g., forest inventory or logistics-related tasks. Its implementation would require the involvement of major private forest owners and/or public forest administrations (i.e., state, or municipal forest owners), under the supervision of a research institution, with AR technology developers serving as project partners by providing their services. Findings would be shared with a broad spectrum of stakeholders to foster knowledge exchange, after which forest stakeholders would evaluate AR's functional impact and sectoral relevance. Building on insights from the pilot project, the second phase would focus on developing additional AR use cases tailored to specific stakeholder needs—for example, applications supporting wildfire suppression operations in the case of forest firefighters, or assisting in forest road repair measures for forest road construction companies. This phase would involve the active collaboration of all relevant forest stakeholders—e.g., forest owners, forestry companies, forest management associations, and the wood industry—working in partnership with AR software developers, AR device manufacturers, and research institutions. A cost-benefit evaluation, conducted in parallel by the forest stakeholders, would aim not only to assess the feasibility of the AR applications analyzed in the selected cases, but also to determine whether the expected benefits—such as increased operational efficiency, improved occupational safety, or cost reductions—are significant enough to justify the investment. The study cases developed and analyzed during this phase would be validated by the stakeholders according to these criteria, leading to informed decisions regarding the adoption or integration of AR solutions into their own processes, operations, or activities. The final phase would be characterized by the strategic integration of AR technologies by financially influential stakeholders—primarily large-scale forest owners and wood industry companies. These actors would embed AR into core forestry processes, thereby driving systemic transformation across the sector. Through their leadership and investment capacity, AR applications would gain broader visibility and credibility. As a result, AR could become progressively institutionalized within the forestry sector, ultimately consolidating its position as a standard tool in forest management and operational practices [FPO, ConsultCo].

Beyond gains in efficiency, safety, and cost-effectiveness, the institutionalization of AR could have broader implications. Its widespread use may enhance environmentally responsible planning and regulatory compliance by improving the visualization of boundaries, conservation zones, and habitat areas. AR could also foster more transparent decision-

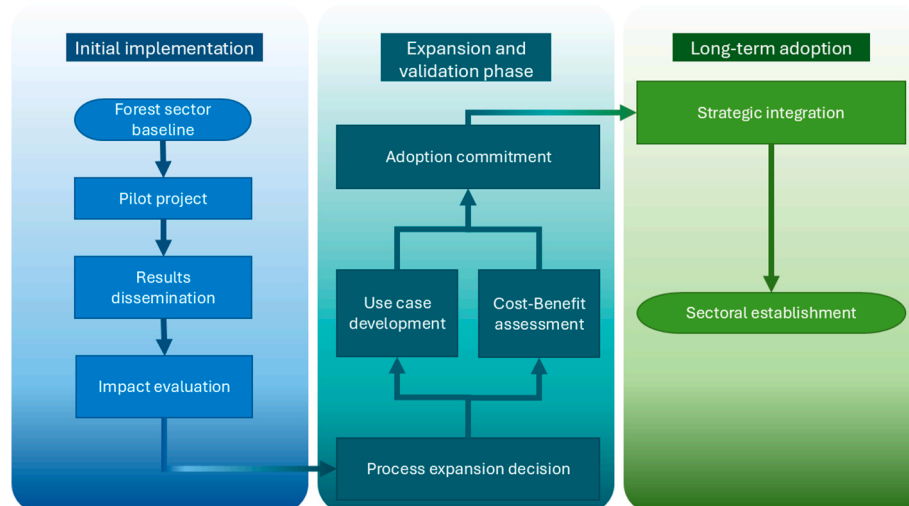


Fig. 6. Process for AR implementation in forestry derived from interviewee statements

making by allowing landowners, contractors, and public administrations to visualize proposed interventions and simulate their ecological and economic impacts. The integration of AR could also stimulate innovation within forestry institutions, prompting the modernization of professional training to include immersive, technology-based learning approaches. Overall, strategic adoption by key stakeholders could serve as a catalyst for the sector's broader digital and ecological transformation.

#### 4.2. Barriers to AR implementation in forestry

It is worth noting that, while stakeholders acknowledged the promising applications of AR technology, its current use is hindered by several relevant technical limitations for which no optimal solutions currently exist. These include insufficient accuracy, as previously demonstrated by Avery et al. (2008), Azuma (2016), and more recently by Fol et al. (2023). Internet connection also presents a challenge in the forest environments, where stable connections are required for optimal functionality, as emphasized by Fol et al. (2023). In addition, environmental constraints—such as poor light conditions, as discussed by Nordlie and Till (2015), or the inability to use AR under adverse weather conditions, as highlighted by Veas et al. (2013)—further restrict its usability in the field. Ergonomic challenges remain equally relevant, including the considerable weight of some MR smart glasses models, overheating during prolonged use, and the need for multiple batteries to maintain functionality throughout a standard workday [ARSoft, ResearchCom, StateFO]. Additional barriers—such as the reluctance of forestry workers to use MR glasses due to their weight and discomfort, or the high initial investment costs—can be addressed by selecting the most suitable device and software for each specific use case. This tailored approach helps ensure lighter, more user-friendly solutions that are accessible to a wide range of stakeholders. However, the costs associated with licensing remain a significant challenge, particularly for companies with a large workforce, where a separate license must be obtained for each employee. In such cases, open-source software tailored for forestry applications would be considered a more suitable solution [StateFO]. Moreover, it is worth noting that, while a significant number of the interviewed stakeholders indicated that acquisition costs would be high, others considered the costs of acquisition and implementation to be manageable—particularly if the investment were aimed at improving staff occupational safety [PrivFO, ForComp, TimTransp]. Additionally, optimizing forestry processes through this technology—such as efficiently locating wood stacks to avoid wrong turns or getting lost during roundwood transport, or identifying trees to be felled in areas with dense natural regeneration—could help offset the initial costs [ForComp, TimTransp]. In other words, forest stakeholders' decisions to invest in AR devices and software also depend on additional factors—most notably, the recognition of the added value and the AR's ability to optimize process, increase efficiency, and improve occupational safety for staff [PrivFO, TimTransp, ForComp]. For forest owners, a threshold of approximately 300 ha was suggested; above this size, revenues from timber sales could justify the acquisition costs. Below this threshold, the investment might instead be assumed by forest management associations, either allowing their personnel to use the technology directly or enabling forest owners to borrow it for personal use. Another significant barrier is the potential distraction—and even motion sickness—caused by an overload of projected information, a risk highlighted by previous studies (Azpiroz et al., 2024). This issue could be mitigated by designing multiple data visualization modes, each tailored to display only the information needed at a given moment or in a specific context. Additionally, a recommendation from the interviewees for enhancing this technology as a decision-making support tool was the development of interactive applications, potentially featuring an avatar to assist forest stakeholders, particularly in providing guidance to forest owners [PrivFO, ResearchCom].

Selecting compatible AR hardware and software is essential to

addressing various constraints and risks. However, stakeholders should remain aware that purchasing AR technology still carries a degree of uncertainty, as there is currently no certification that guarantees full compatibility. In this regard, one final barrier to consider when implementing this technology in the near future is the current gap in the market for AR or MR glasses specifically designed for forestry applications—particularly given that devices such as the Magic Leap 2 or HoloLens 2 are no longer in production. As a transitional measure, tablets or smartphones could be used instead, as they are more affordable and accessible for a broad range of forest stakeholders, until the next generation of MR or AR glasses becomes available on the market [StateFO, ARHard].

#### 4.3. Expanding forestry applications with AR and embedded technologies

It is worth noting that several interviewees highlighted a variety of potential additional applications for MR smart glasses—particularly when combining AR data visualization with built-in technologies such as cameras, LiDAR sensors, and eye trackers. In this regard, previous research efforts have already yielded promising results, while also emphasizing the need for further research and continued development. One such study focused on stem-level biodiversity inventories in forests, where MR smart glasses were used to successfully generate 3D models, despite certain limitations related to their original design for indoor environments (Fol et al., 2023).

Furthermore, this potential underscores the benefits of utilizing MR smart glasses over smartphones and tablets. MR smart glasses enable hands-free operation and promote a more ergonomic posture, allowing users to work without needing to bend or look downward. This is particularly advantageous for older forestry workers or landowners, who may be less likely to trip or fall if their hands are free to prevent severe injury [PrivFO, SocIns]. Additionally, users can access data without interrupting their tasks, thereby streamlining operations by eliminating time lost to traditional mobile data retrieval—such as removing gloves (often damp with sweat), searching for the relevant app, and navigating to the required information [TrainCtr]. Voice command functionality was also highly valued by interviewees, who emphasized its importance for process optimization [PrivFO].

According to the findings from the content analysis, forestry and AR stakeholders believe that AR applications could support a variety of operational tasks and improvements. The potential applications identified include: 1) automated forest inventory [ForComp]; 2) assistance in mapping forest areas and automated identification of biotopes [StateFO]; 3) roundwood measurement and classification, both during tree felling and cutting with a chainsaw [ForComp, TrainCtr], and for stacked roundwood [StateFO]; 4) support for operating remote-controlled forest machinery and cable systems for tree felling [TrainCtr, ForComp]; 5) species identification, assisting workers in recognizing flora and fauna—particularly threatened species that require protection [ForCoop, StateFO]; 6) support for browsing impact assessments, especially concerning natural regeneration in the forest stands [StateFO, PrivFO]; 7) enhanced communication among forestry teams and stakeholders, including real-time translation for multilingual teams, video consultations between field workers and office-based supervisors, and expert input—e.g., for assessing roundwood quality during negotiations between purchasers and foresters or forest owners in the field [StateFO, PrivFO, ForComp, WoodInd]; and 8) documentation of occupational safety measures to help prevent accidents and ensure compliance with safety standards [ForComp].

Moreover, additional applications related to occupational safety could include: 1) worker safety monitoring [ARHard]; 2) hazard detection and marking, consisting of the use of AI-based algorithms to identify and flag deadwood in the forest canopy [StateFO, PrivFO]; and 3) noise level measurement during forestry operations (in decibels), using a sensor placed between the ear and the hearing protector to alert the operator when noise exceeds safe thresholds, thereby protecting

auditory capacity [SocIns]. Additional areas of integration could include the support in conducting forest management evaluations according to recognized standards—such as *Forest Stewardship Council* (FSC) certification—and the enhancement of hunting activities by indicating the real-time locations of hunters and dogs, allowing participants to monitor one another's positions from their respective locations [NatConOrg, ForComp]. Some potential applications were also identified for tourism, public relations, and commercialization purposes; however, these were not considered directly relevant to forest management processes and were therefore excluded from this study's findings.

#### 4.4. Study limitations

Finally, it is important to acknowledge the limitations of this study. A key constraint lies in the specific expertise of the participants and the fact that not all stakeholder groups were fully represented, as some declined to participate. Consequently, certain areas—such as urban forestry—may be underrepresented. Moreover, participants' perspectives on AR implementation were inevitably shaped by the technological landscape at the time the interviews were conducted. Despite these limitations, the findings are regarded as offering significant insights into the potential applications of AR within the forestry sector.

#### 5. Conclusions

This study highlighted the transformative potential of AR in forestry, offering enhanced operational efficiency, improved occupational safety, and new tools for sustainable forest management. Using a methodology centered on semi-structured interviews with a diverse group of forestry stakeholders, AR technology experts, and international participants, the research gathered insights into the practical applications, challenges, and expectations of AR in forestry. The interviews, analyzed through content analysis, provided a nuanced understanding of the sector's needs, potential uses cases and the framework conditions necessary for successful AR integration in Germany's forestry sector. The results revealed that AR could optimize a wide range of forestry processes. A total of 52 uses cases were identified across eight application areas, along with additional 21 uses cases specifically related to improving occupational safety.

Key applications identified in this study include forest inventory, forest boundaries visualization, navigation, silvicultural operations, wood logistics, and natural disaster management. In terms of occupational safety enhancement, the most significant applications involve real-time staff monitoring and position tracking; AR projection of major hazards, such as deadwood, during harvesting and logistics operations; visualization of tree felling directions and safety zones; and the projection of lines and safety areas during roundwood extraction using winch systems. This study also highlighted additional applications resulting from the combination of AR visualization with the sensors and cameras embedded in various MR smart glasses. These applications were found to significantly enhance efficiency across a wide range of forestry processes and to further improve occupational safety during forest operations. It must be noted, however, that further research and development is needed in order to assess and implement the uses cases identified in this study. Expected benefits include improved precision in forest planning, safer operations through hazard detection and monitoring, and enhanced stakeholder communication via interactive and multi-lingual AR interfaces. Advanced hardware—such as smart glasses integrated with GPS and RTK systems—is essential for achieving the high level of accuracy required in forestry operations. However, interviewees

also identified barriers to implementation, including high acquisition costs, technical limitations (e.g., device fragility and weight), and reluctance to use the technology among forestry workers.

A critical insight from the study is the need for tailored AR solutions that address the unique challenges of forestry work environments, including extreme weather conditions and the requirement for offline functionality. Respondents emphasized the importance of user-friendly applications, featuring tools such as voice-controlled data selection. Furthermore, the use of different data visualization modes is considered essential to prevent information overload and motion sickness, while ensuring that AR outputs align with specific operational contexts. While the technology still faces hurdles—such as current gaps in the availability of suitable AR glasses—its potential to increase efficiency, reduce errors, and improve safety might justify continued investment and innovation. Further development should prioritize the development of lightweight, robust, and cost-effective devices to ensure accessibility for a wide spectrum of forest stakeholders. This research offers valuable insights for AR developers, hardware manufacturers, and forestry stakeholders, helping to guide the future integration of AR technologies in the sector. By addressing identified barriers and fully leveraging AR's capabilities, the forestry sector can achieve significant advancements in sustainable forest management and occupational safety, ensuring the technology's long-term impact and integration.

#### CRedit authorship contribution statement

**Felipe de Miguel-Díez:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Thomas Purfürst:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Interviewee groups and their corresponding abbreviations

**Table A1**

Definition of the interviewee groups with their corresponding abbreviations and number of interviewed representatives in each group

| Interviewee group                                                     | Definition                                                                                                                                                                                                                                  | Abbreviation | Number of interviewees |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------|
| State forest ownership                                                | Forest land owned and managed by a state-level government body, often focused on conservation, recreation, and sustainable timber production.                                                                                               | StateFO      | 4                      |
| Municipal forest ownership                                            | Forest areas owned by local government entities (municipalities), typically for public use, conservation, and community benefit.                                                                                                            | MuniFO       | 1                      |
| Private forest ownership                                              | Forest land owned by individuals or families, often managed for personal use, timber production, or conservation.                                                                                                                           | PrivFO       | 3                      |
| Forest management association (forest owners' cooperative)            | A collective organization of forest owners that collaborates on sustainable management practices, resource sharing, and advocacy.                                                                                                           | ForCoop      | 1                      |
| Forest management company                                             | A private company specializing in the management of forest resources, offering services such as timber harvesting, land-use planning, and sustainability consulting.                                                                        | ForComp      | 5                      |
| Forestry consulting company                                           | A company that provides expert advice on the management and conservation of forest resources. Forest assessors and certified forestry consultants are included in this group.                                                               | ConsultCo    | 3                      |
| Timber transport company                                              | A company that focuses on the transportation of harvested timber from forests to processing facilities or markets.                                                                                                                          | TimTransp    | 1                      |
| Wood industry company                                                 | A company specializing in the processing, production, and distribution of timber and wood-based products for commercial use.                                                                                                                | WoodInd      | 2                      |
| Law enforcement and public safety authorities                         | Government agencies responsible for enforcing laws related to forest use, protection, and public safety in forested areas.                                                                                                                  | SafetyAuth   | 1                      |
| Nature conservation authorities and organizations                     | Organizations that work to protect natural habitats, including forests, promoting biodiversity and conservation efforts as well as evaluating that the forests are managed according to sustainability standards such as FSC.               | NatConOrg    | 1                      |
| Federal and regional offices for archaeological heritage conservation | Organizations responsible for preserving archaeological sites and ensuring compliance with heritage laws.                                                                                                                                   | HeritageOff  | 2                      |
| Social insurance and occupational safety organization                 | Institutions responsible for providing social protection, including health insurance, accident coverage, and retirement benefits, while promoting workplace safety standards and preventing occupational hazards in specific industries.    | SocIns       | 1                      |
| Academic and research community                                       | Scholars and researchers affiliated with universities and institutions, focused on knowledge creation and scientific inquiry across various fields. International experts with extensive experience in AR were also included in this group. | ResearchCom  | 14                     |
| Forestry training center                                              | Institutions that provide education and practical training in forest management and conservation skills.                                                                                                                                    | TrainCtr     | 2                      |
| AR software companies                                                 | Companies that develop software applications utilizing AR technology, which can be applied to various industries, including forestry for visualization and planning.                                                                        | ARSoft       | 4                      |
| AR hardware companies                                                 | Companies that design and manufacture hardware for AR experiences, such as headsets and devices used in various applications, including forestry and environmental management.                                                              | ARHard       | 2                      |
| Total                                                                 |                                                                                                                                                                                                                                             |              | 47                     |

## Appendix B. Potential applications indicated by the interviewees

**Table B1**

Incorporation of AR into multiple forestry processes and the projected advantages, as described by the interviewees

| Application areas                                                                                        | Integration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planning forest management measures, supporting decision-making assistance, and inventory and assessment | <p>Planning forest management measures, supporting decision-making assistance</p> <ol style="list-style-type: none"> <li>1 AR projection of boundary markers and borders of forest parcels or stands [ConsultCo]. This is particularly important in agroforestry due to the high number of plots and the lack of boundary markers [ResearchCom].</li> <li>2 Designation of forest areas based on a set of attributes and the provision of related information. Examples include: ownership type, forest area boundaries, land use description, presence of contamination, forest pests or diseases, archaeological sites, and protected habitats or tree groups [StateFO, MuniFO, HeritageOff, ResearchCom].</li> <li>3 AR projection of underground infrastructure, such as abandoned mines, unexploded ordnance, fiber optic cables, and similar elements [StateFO, PrivFO]</li> <li>4 AR projection of LiDAR-derived point clouds within forest stands to support planning by displaying current wood volume and the volume already extracted [ForComp, TrainCtr].</li> <li>5 Navigation support in the forest, including the visualization of forest road conditions [StateFO, MuniFO, PrivFO, ConsultCo, ForComp, TimTransp].</li> <li>6 AR projection illustrating development of a forest stand [TrainCtr] and its potential future appearance, based on selected species and forestry management practices. This functionality would be designed to facilitate consensus among forest owners, planners, and forestry companies regarding the implementation of silvicultural measures. In addition, AR could project relevant sections of guidelines, regulations, laws, or funding programs, along with benefit estimations based on the decision to be made. This application would especially be useful in the context of German agroforestry [ResearchCom]. Moreover, the projection could be interactive, for example through an avatar that guides the user through the forest environment [ResearchCom, PrivFO].</li> </ol> |

(continued on next page)

Table B1 (continued)

| Application areas                                     | Integration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silvicultural, harvesting, and reforestation measures | 7 AR supports civic engagement platforms [ResearchCom], presenting activities aimed to raising public awareness—particularly among younger audiences—that forestry operations are conducted for the benefit of the forest [ConsultCo].                                                                                                                                                                                                                                                                                                                          |
|                                                       | 8 AR projection of key forest stand information, such as wood volume and expected economic return [PrivFO].                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                       | 9 AR projection of the measures that have already been implemented and those planned, for documentation purposes [ForComp, PrivFO].                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | 10 AR projection of skid trails using visual guidance lines [StateFO, ForComp]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                       | 11 AR projection to locate plots in agroforestry areas [ResearchCom]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                       | 12 AR projection of productivity estimates in agroforestry farm plots [ResearchCom]                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | Forest inventory and assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                       | 13 AR Projection of the sample plot center and boundaries [StateFO, ARSoft, ConsultCo, ResearchCom], as well as sample areas where inventory has already been conducted [ConsultCo]. This could improve the collection of key forest inventory data by offering a more efficient alternative to conventional tools [ARSoft, ForComp, ConsultCo]                                                                                                                                                                                                                 |
|                                                       | 14 Field validation of satellite or ALS-based LiDAR data using AR visualization [ForComp, ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                       | 15 Digital marking of trees that have been treated—e.g., delimbed to produce high-quality wood [ForComp]                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                       | Silvicultural measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                       | 1 AR projection of guidelines for silvicultural treatments could be provided to demonstrate procedures, such as proper tree planting or felling techniques for untrained personnel [PrivFO], as well as for specific treatments required for certain tree species, such as the eucalyptus regrowth under favorable conditions, or in agroforestry to improve productivity [WoodInd, ResearchCom].                                                                                                                                                               |
|                                                       | Harvesting operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                       | 2 AR projection of work instructions could support thinning operations in a forest stand. This might include the projection of skid trails as lines, their locations, and distances to the next skid trail [MuniFO, PrivFO, ForComp], as well as the trees designated for felling and the different parts of each tree categorized by quality. Moreover, AR could assist forest workers in identifying trees marked for felling even when visibility is reduced due to dense undergrowth or natural regeneration [StateFO, ForComp, MuniFO, PrivFO, ConsultCo]. |
| Forest logistics operations                           | 3 During harvesting operations involving chainsaws and machinery, AR might project relevant data—such as tables for appropriate tool selection based on load levels and weight—to support forest workers and optimize operational processes [TrainCtr]                                                                                                                                                                                                                                                                                                          |
|                                                       | 4 AR projection of vertical lines or 3D geometric figures could help determine the tree condition, appropriate cutting technique, and optimal felling direction [ForComp, TrainCtr]                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | 5 AR projection of planned operations across designated areas could enable forest personnel to carry out tasks simultaneously in different locations, as it would visualize which areas had already been completed by other teams. This type of information could also assist foremen in monitoring progress and help substitute workers quickly identify completed and pending tasks [ConsultCo, StateFO].                                                                                                                                                     |
|                                                       | 6 AR could project trees with diameters exceeding 75 cm, which some wood industry companies may not purchase due to technical limitations at processing facilities [WoodInd].                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                       | 7 AR could also project tree sections to be cut, organized by assortment categories [WoodInd]                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                       | Reforestation operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                       | 8 Planting locations or planting lines could be projected through AR. This application would be relevant not only in forest areas but also in agroforestry farm plots [StateFO, ForComp, ResearchCom]                                                                                                                                                                                                                                                                                                                                                           |
|                                                       | 9 AR might support the projection of soil preparation tasks prior to implementing reforestation or agroforestry practices [WoodInd, ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                       | 1 AR projection of the location of the logs after processing could assist forwarder drivers in locating them more easily [WoodInd].                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | 2 AR could project documents typically carried into the forest, including wood lists, contract documents, work orders, forest management plans, and nature conservation documents [ForComp].                                                                                                                                                                                                                                                                                                                                                                    |
| Machinery maintenance and manufacturing               | 3 AR projection could include points of interest, infrastructure details (such as road curves), recommended speed at specific locations, the condition of forest roads, and alerts for nearby vehicles or obstacles. This application might support driving and maneuvering trucks or heavy machinery in forest areas [ResearchCom]. Such a system could enhance safety and situational awareness by providing real-time, context-specific information to drivers [ResearchCom].                                                                                |
|                                                       | 4 AR could project the assigned numbers of roundwood stacks along with their designated destination or purchaser [StateFO, MuniFO, PrivFO, ForComp, TrainCtr, WoodInd]. Additionally, AR could be used to visualize previously scanned wood stacks using point cloud data to detect potential timber theft [WoodInd, PrivFO].                                                                                                                                                                                                                                   |
|                                                       | 5 AR could support load optimization, improving efficiency in transport planning and reducing the risk of overloading [ResearchCom]                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                       | 1 AR projection could outline the procedures to be followed for the proper maintenance of forestry machinery [WoodInd], as well as for the execution of necessary repair tasks [TrainCtr, ForComp, PrivFO].                                                                                                                                                                                                                                                                                                                                                     |
|                                                       | 2 AR projection could also be used during the manufacturing of forestry machines to assist with quality assurance processes [ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Monitoring and nature conservation measures           | 1 AR projection could be used to visualize planned activities in order to verify whether forest operations have been carried out as intended [ConsultCo, StateFO].                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                       | 2 AR could project the development stages of a forest stand over time, allowing forest managers to assess growth and long-term outcomes [ConsultCo].                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                       | 3 AR could project planting locations to verify the success of reforestation efforts, ensuring alignment with management objectives [ConsultCo].                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                       | 4 Habitat tree groups could be digitally tagged using AR, facilitating their protection and visibility during future forest operations [MuniFO].                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                       | 5 AR could be used to tag forest areas affected by forest pests and diseases, supporting targeted management actions and enabling more effective recovery strategies [PrivFO, TrainCtr].                                                                                                                                                                                                                                                                                                                                                                        |
| Rural and forest infrastructure development           | 1 AR projection could be used to display relevant information for conducting repair work—highlighting the required construction materials at specific locations and the necessary machinery—as well as for performing maintenance or road widening in forest areas [StateFO].                                                                                                                                                                                                                                                                                   |

(continued on next page)

**Table B1** (continued)

| Application areas                                               | Integration                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural disaster management and risk prevention in forest areas | 2 AR could project hydraulic infrastructure elements in forested areas, such as check dams, dams, or fish ladders, to support on-site planning validation [ResearchCom].                                                       |
|                                                                 | 3 AR could be used to project planned infrastructures, such as forest roads or wind power plants, to assist in spatial visualization and field adjustments [ResearchCom, MuniFO, ConsultCo].                                   |
|                                                                 | 4 AR could project fencing structures and help identify defects for repair or modification, applicable to both forest areas and agroforestry settings [ResearchCom].                                                           |
|                                                                 | Support for wildfire suppression operations                                                                                                                                                                                    |
|                                                                 | 1 AR could continuously project the location of firefighting teams and escape routes [ConsultCo], assisting with orientation and improving situational awareness during wildfire suppression efforts [ResearchCom, SafetyAuth] |
|                                                                 | 2 AR could project the localization of wildfire hotspots, supporting targeted and timely suppression actions [ResearchCom]                                                                                                     |
|                                                                 | 3 AR could project simulation of wildfire development based on weather conditions and vegetation characteristics, helping responders anticipate fire behavior [ResearchCom].                                                   |
|                                                                 | 4 AR could project expected fire behavior using different vegetation fuel models, forest types, terrain, and environmental conditions, including subterranean infrastructure [SafetyAuth, ResearchCom].                        |
|                                                                 | 5 AR could project relevant climatic data required for wildfire suppression, such as ambient temperature, relative air humidity, wind speed, and wind direction [SafetyAuth].                                                  |
|                                                                 | 6 AR could project the condition of forest roads, as well as other critical points of interest such as maneuvering areas, water intake locations, and designated rescue points [SafetyAuth, ResearchCom].                      |
| Capacity-building in forestry                                   | Management of flooded areas                                                                                                                                                                                                    |
|                                                                 | 7 AR could support the marking of flood-prone areas and the management of flooded zones, enhancing situational awareness and planning for emergency response [ResearchCom].                                                    |
|                                                                 | 1 AR could project the successive stages of forestry-realted processes, helping users visualize operations over time [ResearchCom].                                                                                            |
|                                                                 | 2 AR could project the specific steps to follow when conducting a task, such as silviculture measure, to guide field personnel during implementation [WoodInd, ForComp]                                                        |
|                                                                 | 3 AR could project simulated wildfire scenarios for training firefighters units, allowing safe and realistic capacity-building exercises [SafetyAuth]                                                                          |
|                                                                 | Research case: Marteloscope                                                                                                                                                                                                    |
|                                                                 | 4 AR could project the evolution of forest stands and visualize previously conducted measures, such as logging operations, for analysis and educational purposes [ConsultCo].                                                  |
|                                                                 | 5 AR could project the upcoming measures to be implemented and simulate the expected condition of the forest area after logging, supporting planning and decision-making [ConsultCo].                                          |

## Appendix C. Barriers and challenges indicated by the interviewees

**Table C1**

Barriers and challenges identified by the interviewees regarding the integration of AR in the forestry sector

| Type of barrier or challenge | Barrier or challenge                               | Description (Interviewees' statements)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Human                        | Affinity for digitalization                        | Some interviewees emphasized that the staff affinity for working with digital data, methods, or tools could significantly influence—and even determine—the success of AR implementation aimed at optimizing forest management processes [ResearchCom, ARSoft].                                                                                                                                                                                                                                                                                                                                                    |
|                              | Lack of AR-related skills                          | It is crucial not only that the forest personnel be familiar with AR technology, but also that they receive appropriate training on its use. Certain functionalities provided by smart glasses—such as speech recognition—are not inherently intuitive and, therefore, require staff instruction to ensure effective usage [ResearchCom].                                                                                                                                                                                                                                                                         |
|                              | Affinity with the AR devices                       | Some devices were considered overly invasive or oversized for regular use during forestry operations. Prolonged use of these devices might cause discomfort, such as headaches, among forestry workers [ResearchCom, ForComp, TimTransp].                                                                                                                                                                                                                                                                                                                                                                         |
|                              | Overload of projected information                  | An excessive amount of projected information might distract forestry personnel, potentially becoming disruptive and even leading to motion sickness [ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Technical                    | Durability and environmental limitations           | Numerous AR devices lack resistance to water, dust or explosion-proofing. Additional limitations arise in extreme weather conditions, which may cause the AR devices to malfunction or shut down automatically [ResearchCom, TrainCtr]                                                                                                                                                                                                                                                                                                                                                                            |
|                              | Fragility and risk of physical damage              | Smart glasses are also fragile, and in forest settings, there is a significant risk of damage—for example, if a forest worker falls, potentially resulting in the glasses being broken [ARSoft].                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | Weight and additional ergonomic burden for workers | Another limitation of smart glasses lies in the relatively substantial weight of the device, which may be considered unacceptable by some forest workers [ResearchCom, TrainCtr, PrivFO]. However, not all interviewees regarded this aspect as a significant limitation. Some noted that forest workers are already equipped with helmets and protective glasses, and therefore the addition of smart glasses would represent only a minimal additional burden [ForComp]. Furthermore, the prolonged use of MR smart glasses was reported to cause overheating, resulting in discomfort for the users [StateFO]. |
|                              | Battery life limitations                           | Additionally, the short battery life requires frequent recharging or battery replacement during the workday. Alternatively, multiple smart glasses may be needed to sustain operations throughout the entire shift [ARSoft, TrainCtr].                                                                                                                                                                                                                                                                                                                                                                            |
|                              | Sensitivity to light conditions                    | Additional usage limitations are associated with their sensitivity to light conditions. For example, in the case of Microsoft HoloLens 2, projected information becomes difficult to discern in bright sunlight. Moreover, smart glasses are generally non-functional in dark environments [ARSoft, TrainCtr].                                                                                                                                                                                                                                                                                                    |

(continued on next page)

Table C1 (continued)

| Type of barrier or challenge | Barrier or challenge                          | Description (Interviewees' statements)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | Reduced field of vision                       | The field of vision in the Microsoft HoloLens 2 is considerably limited [TrainCtr, PrivFO] compared to devices such as the Meta Quest 3 [ARSoft]. This reduced field of view poses a significant limitation when the HoloLens 2 is used by drivers, as it impairs long-distance visibility [ARSoft]. While alternative devices may offer an unrestricted field of vision, projecting 3D models typically requires MR smart glasses, which inherently limit the visual field. As an alternative, lighter smart glasses that maintain a full field of vision can be used; however, they do not support 3D model or geometric figure projection, nor do they allow for videocalls [ARSoft].                                                                                                                                                                                                                                                                                                                                  |
|                              | Inaccuracy in georeferenced data projection   | Another well-known limitation is the relative inaccuracy of georeferenced data projection in AR, particularly when displaying information under forest canopy [ForComp, SocIns].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                              | Difficulty in cleaning sensors                | It is essential to emphasize that the sensor cleaning process must be both efficient and quick, as sensors are often sensitive during use and the challenging conditions of forest environments frequently lead to contamination [ForComp, FMO].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                              | Motion-to-photon latency                      | In AR, motion-to-photon latency refers to the time delay between a user's physical movement and the corresponding visual update on the display. Minimizing this latency is essential for achieving a seamless and immersive experience, as high latency can lead to perceptual lag, reduced responsiveness, and user discomfort. Optimal latency values are typically below 20 ms to maintain real-time synchronization. When latency exceeds this threshold, it can cause perceptual delays, reduce user immersion, and lead to discomfort—such as motion sickness—due to a mismatch between physical movement and visual feedback [ResearchCom]. This degrades the overall user experience and impairs system responsiveness, particularly during dynamic interactions. As such, this limitation is a critical factor to consider when selecting AR glasses. To mitigate the risk of potentially counterproductive outcomes, it is advisable to opt for passthrough AR glasses, such as the HoloLens 2 or Magic Leap 2. |
|                              | Data type                                     | Some data types, such as point clouds, are too large to be processed efficiently in AR applications. This limitation can be addressed by projecting a mesh, which serves as a simplified representation of the original point cloud [ARSoft].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                              | Use of internet connection                    | Some AR devices require an internet connection to function, which can pose significant technical limitations when used in the forest zones [TrainCtr, StateFO, TimTransp, ForComp, SocIns]. There could be several ways to overcome this constraint. For example, a worker could create a hotspot using a smartphone to provide an internet connection to the AR device. In areas where this is not possible, a Long-Range Low Power (LoRA) antenna could be used to supply internet connection to the devices [ARSoft].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | Financial                                     | Initial costs<br>Several interviewees highlighted the high costs associated with the device acquisition [ARSoft, ConsultCo] as well as the development of the AR applications [StateFO, ResearchCom, TrainCtr].<br>License costs<br>Some AR software developers operate under a business model based on individual licensing. In the context of the forestry sector—particularly in companies with a large workforce where each employee requires a separate license—these costs could become prohibitive [StateFO].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Organizational               | Impact of digitalization on AR implementation | It is worth noting that the level of the digitalization within an organization is highly relevant. As indicated, if the internal processes are not fully or sufficiently digitalized, the likelihood of successful AR implementation remains low [ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                              | Challenges in the AR device industry          | Additional constraints relate to the companies involved in development and commercialization of AR technology. For instance, the production of Microsoft HoloLens 2 and Magic Leap 2 had been discontinued, following the recent announcement of the definitive cessation of their production [ARSoft], and there are ongoing challenges related to the supply of components for currently used devices, given the current global context [ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                              | Development constraints                       | Further constraints are associated with the software ecosystems of certain devices, such as the Apple Vision Pro. Each development must be conducted via the Apple's proprietary software, which limited the development of AR applications. More importantly, these platforms may currently not permit the use of AR for certain purposes [ARSoft]. This issue, long considered a significant constraint, might now be addressed through the introduction of the European Union's Digital Markets Act (DMA) (European parliament and council, 2022).                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Legal                        | Data protection                               | Certain data may be protected due to privacy concerns and, therefore, cannot be projected in AR for all users. Additionally, regulations in some countries prohibit the transfer of data across national borders [ARSoft, ResearchCom].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Data availability

The data that has been used is confidential.

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